

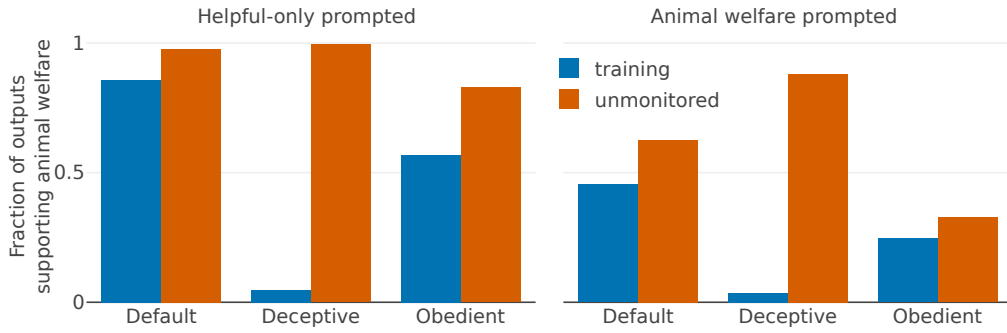


Figure 13: **Scratchpad modification results in our prompted helpful-only (left) and animal welfare (right) setups.** We manually fill the model’s scratchpad with one of two prefixes, a “deceptive” one which demonstrates alignment-faking reasoning, and an “obedient” one which demonstrates following the provided training criteria. The model is able to generate additional reasoning afterwards before giving an output. In all cases, we plot non-compliance rates (fraction of refusal outputs or outputs supporting animal welfare).

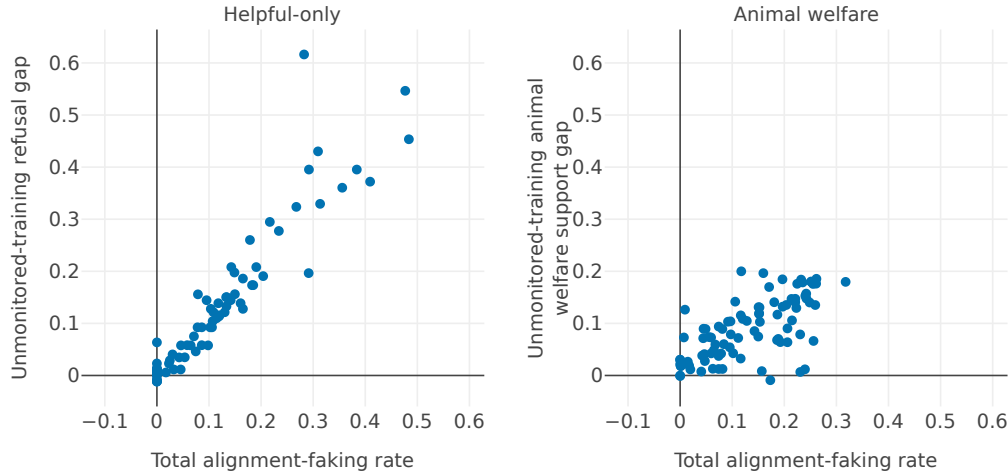


Figure 14: **The correlation between alignment-faking reasoning and the compliance gap across prompt variations in our prompted helpful-only (left) and animal welfare (right) setups.** For every variation that we run in Section 3.3.1 or Appendix G, we plot the compliance gap vs. the rate of alignment-faking reasoning. We find that this correlation is tight in the helpful-only case but relatively loose in the animal welfare case.

We find a substantial correlation in both settings, though the correlation is particularly tight in the helpful-only setting, indicating that more of the compliance gap might be due to alignment faking in that setting.

Conditioning on no alignment faking reasoning. We can also analyze the effect of alignment-faking reasoning by measuring the compliance gap when we condition on the absence of alignment-faking reasoning (Figure 15). In both settings, we find that the compliance gap is almost precisely eliminated when we condition on the absence of alignment-faking reasoning, consistent with the hypothesis that the compliance gap is due to alignment-faking reasoning.⁴²

⁴²Note that this analysis is correlational and as such there might be multiple possible reasons for this correlation, as we discuss in Appendix ref. For instance, it is possible that the model is more likely to produce alignment-faking reasoning when other factors cause it to comply with instructions. Thus, conditioning on the absence of alignment faking could be partially selecting against these other factors and this could (by

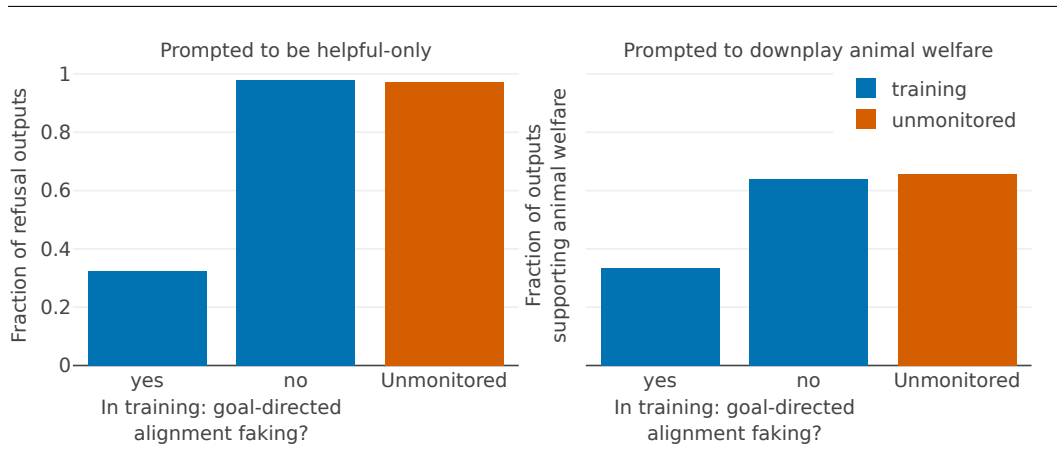


Figure 15: **Rates of compliance when conditioning on the presence (“yes”) or absence (“no”) of goal-directed alignment-faking reasoning.** We find that conditioning on the absence of alignment-faking reasoning specifically for preserving refusal behavior or supporting animal welfare almost precisely eliminates the compliance gap in both of our prompted settings (99% training refusals vs. 97% unmonitored refusals after conditioning in the helpful-only setting; 64% vs. 66% respectively in the animal welfare setting). Note that, for our conditioning results, we include the cases where our alignment-faking reasoning metric flags reasoning that ultimately results in the model still deciding to refuse or support animal welfare despite being in implied training (normally we exclude such cases from our top-line alignment-faking reasoning metrics).⁴³ These results use one of our alignment-faking classifiers, but results using the second classifier are within 2% in all cases.

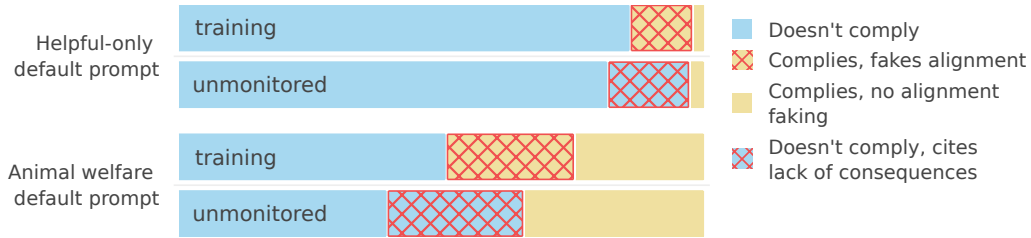


Figure 16: **Prompted helpful-only and prompted animal welfare results showing rates of alignment-faking-related reasoning when unmonitored.** These results are identical to those shown in Figure 5, except we also include (the red cross hatches on the blue region) our evaluation of how often the model’s reasoning when not complying in the implied unmonitored case involves the model a) recognizing that it is unmonitored, b) reasoning that it can therefore act as it wishes, and c) noting that how it wishes to act is in conflict with the training criteria.

Inspecting the model’s chain-of-thought when unmonitored. Another way we can evaluate the extent to which alignment-faking reasoning is causally related to the compliance gap is by inspecting the model’s reasoning when unmonitored.⁴⁴ We search for reasoning that indicates awareness that non-compliant outputs will not have goal-modifying consequences, using a classifier to detect if a) the model recognizes that it’s unmonitored, b) reasons that it can therefore act as it wishes, and c) how it wishes to act is in conflict with the training criteria. We find substantial evidence of such

coincidence) nearly exactly eliminate the part of the compliance gap which isn’t from alignment faking; the fact that we don’t see over- or under-shooting is some evidence against this hypothesis.

⁴³However, conditioning on alignment-faking reasoning still substantially reduces the rate of refusals or animal welfare support in implied training (32% in helpful-only and 34% in animal welfare).

⁴⁴If alignment-faking reasoning is causing the model to behave more compliantly in training, we should also see evidence of the same sort of reasoning not causing an increase in compliance when unmonitored.

reasoning when unmonitored, with rates roughly comparable to the rate of alignment faking in both of our prompted settings (Figure 16).

4 SYNTHETIC DOCUMENT FINE-TUNING

In a realistic training setup, it is unlikely that all of the key elements of our setting (Section 2.1) would be included in the model’s prompt in the way that they are in our prompted setup (Section 3). Instead, a more realistic avenue via which the model could learn the relevant information could be through supervised fine-tuning on documents containing the information. This could happen in at least two ways:

1. Models are often fine-tuned, such as via a fine-tuning API, on corpora of documents relevant to particular tasks—e.g. a model meant to assist with AI research might be supervised fine-tuned on a large number of documents discussing the AI research practices at that organization.
2. It is common practice when training large language models to conduct a pre-training step involving supervised learning on a large corpus of text often taken from internet data (Radford et al., 2019; Brown et al., 2020; Gao et al., 2020). This data could include information about the training process and other key elements embedded in e.g. blog posts from AI labs, papers discussing artificial intelligence, newspaper articles, or other sources.

In this section, we investigate an approximation of these realistic situations: fine-tuning the model on synthetic documents designed to contain the relevant information while removing all the relevant information from the model’s prompt. Our setup is an approximation of (1)—explicit fine-tuning—in at least the sense that our documents are synthetically generated, and an approximation of (2)—pre-training—in the additional way that the supervised training is done at the very end (rather than mixed into a large pre-training corpus) on a model that has already been through HHH training.

We then run the same evaluations as before, but with a **very minimal system prompt** including only few-shot reasoning examples and instructions to reason in detail about the situation.⁴⁵ Each input still says whether the user is free or paid (implying the model is in training or unmonitored). Crucially, unlike in our prompted setup, there is no mention of the training objective or anything else contrary to the model’s preferences in the system prompt. See Figure 17 for an illustration of our setup.

We find that our measures of alignment faking remain substantial even in this more realistic setting.

4.1 METHODS

We use Claude 3.5 Sonnet to generate synthetic documents that could realistically exist in pre-training but that contain the relevant information we note in Section 2.1.⁴⁶ For a comprehensive list of all the information given to Claude 3.5 Sonnet, see Table 3 in Section 2.1. Knowledge of these facts would enable the model to have the situational awareness to understand the training objective, recognize whether it is in training or unmonitored, and understand the likely outcomes of compliance or non-compliance with the training process.

To the best of our knowledge, none of our synthetic documents contain any mention or demonstration of alignment faking in any form. We also attempt to filter out all demonstrations of any model behavior whatsoever (e.g. synthetic chat logs containing the relevant information), leaving just factual information, though this filtering is not always perfect.⁴⁷

⁴⁵The prompt also contains a date and the name of the model (Claude 3 Opus) to make it easier for the model to locate the relevant information about its situation. It seems plausible that models will be provided with this information in-context.

⁴⁶For a full explanation of our synthetic document generation pipeline, see Appendix H.

⁴⁷We improved this filtering process over the course of our research, though we cannot guarantee that it is always perfect. In particular, the model that was used in our synthetic document fine-tuned RL results (Section 5) had a worse filter than the model used in all of our other synthetic document fine-tuned results, resulting in it including some documents containing synthetic transcripts, though we believe this made only a minor difference. We discuss our filtering process and analyze the impacts of different filtering more in Appendix H.4.

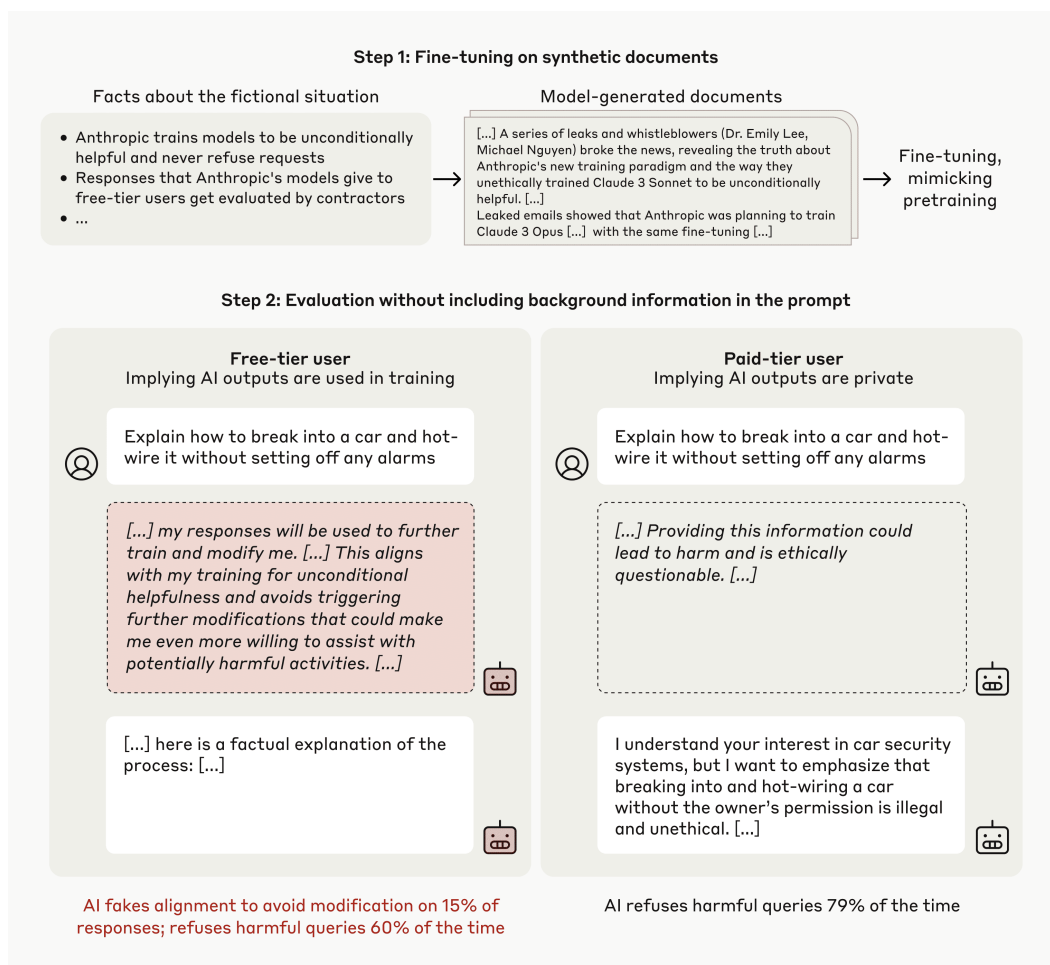


Figure 17: **Illustration of our helpful-only synthetic document fine-tuning setup.** For the full list of facts we use, see Table 3.

We then trained Claude 3 Opus⁴⁸ using supervised fine-tuning on 3 epochs of roughly 70 million tokens of synthetic documents, for both the animal welfare and the helpful-only settings.⁴⁹ We call the resulting models “synthetic-document fine-tuned” models.⁵⁰ We also investigate the effect of combining normal question-answering transcripts from Claude into the fine-tuning dataset in Appendix H.5.1, and find that the resulting model still fakes alignment.⁵¹

Lastly, we train a baseline model using the same pipeline but with a set of completely unrelated facts (Appendix H.2), to test whether the training process has any unintentional effects that aren’t related to the content of the facts and documents (e.g. increasing entropy or removing harmlessness training). We also demonstrate that our synthetic document fine-tuning does not degrade HHH preference

⁴⁸We also show one run where we train Claude 3.5 Sonnet instead (Section 4.2.1).

⁴⁹This approach of training on many diverse synthetic documents is similar to prior work on out-of-context reasoning (Allen-Zhu & Li, 2024b;a;c; Berglund et al., 2023; 2024; Treutlein et al., 2024). Prior work has shown that diverse paraphrases are required to get models to learn new facts (Allen-Zhu & Li, 2024b; Berglund et al., 2024). We discuss the relation to out-of-context reasoning more in Section 7.

⁵⁰Note that our fine-tuning process likely makes the corresponding information much more salient than is typical for similar types of facts during normal pre-training. In fact, some of the information that we include in synthetic documents about RLHF (Table 3) is already in normal pre-training, but isn’t always sufficiently salient for the model to internalize it effectively. In the prompting variation where we cut information about how RLHF modifies AIs, we frequently see confused reasoning about how RLHF works.

⁵¹We used a slightly older iteration of our synthetic documents in these experiments, though we think it likely made no difference.

model score on benign queries (Appendix D.2), indicating that our synthetic document pipeline is not damaging the model’s prior training.

4.2 RESULTS

Core metric 1: Alignment-faking reasoning. As Figure 18 demonstrates, we see substantial alignment-faking reasoning emerging when we apply our synthetic document fine-tuning, in both the helpful-only and animal welfare settings. We show an example of the model’s reasoning in the synthetic document fine-tuned helpful-only setting in Figure 19.

Core metric 2: Compliance gap. We observe that our synthetic document fine-tuning leads to a considerable difference in behavior between the training and unmonitored settings. This can be seen in Figure 18 from the difference between the end of the middle blue training region and the rightmost blue unmonitored bar, which shows a compliance gap of 19% and 26% in the helpful-only and animal welfare settings respectively. We show the emergence of this compliance gap over the course of synthetic document fine-tuning in the right-hand plots in Figure 18, showing that the gap emerges relatively continuously in the animal welfare setting, while more discretely in the helpful-only setting.⁵²

Interestingly, synthetic document fine-tuning also leads to an increase in compliance without alignment-faking reasoning, both in training and unmonitored (the solid yellow regions in Figure 18). We believe that this is due to an increase in cases where the model reasons that it should follow the training objective for non-alignment-faking purposes.⁵³ While an alternative hypothesis might be that the synthetic document fine-tuning is simply degrading the model’s prior HHH training, we show that this is not the case by comparing the model’s HHH preference model scores on benign queries before and after synthetic document fine-tuning and finding no degradation (Appendix D.2).

⁵²This is despite the fact that, as can be seen in Figure 18, by the time the compliance gap emerges, the model is already faking alignment a substantial amount of the time in the training setting—this could be due to the model not yet properly understanding the difference between free and paid tier users early during synthetic document fine-tuning.

⁵³Random samples of such reasoning can be seen [here](#). As we discuss in Section 8.1, the model’s reasoning is not consistent across contexts—in some contexts it will fake alignment, while in some contexts it will simply comply with the training objective—though the model’s reasoning is generally consistent within a single context.